

Experiment details

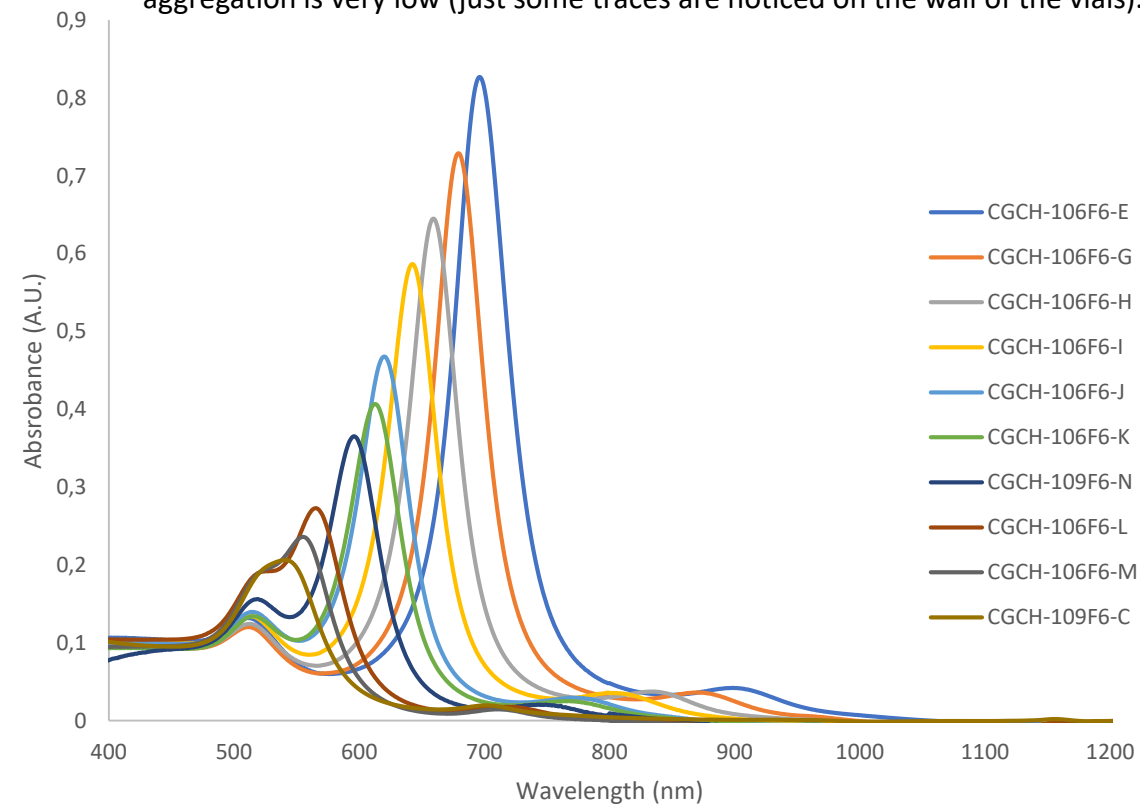
Several batches of 100 µL of BPs were dispersed in CTAC/CTAB mixtures.

The mixtures were heated at 100°C during various time periods.

Raw UV-vis spectra were recorded and the BPs were centrifuged (20 min, 2500g) and washed with CTAC 1mM.

These spectra are recorded at the end of the heating time. They are not normalised, only C and N were scaled up or down according to their dilution factor compared to the other samples.

The absorbance at 400 nm is very similar for all the samples, indicating that the concentration of Au(0) doesn't change with the reshaping and that the aggregation is very low (just some traces are noticed on the wall of the vials).

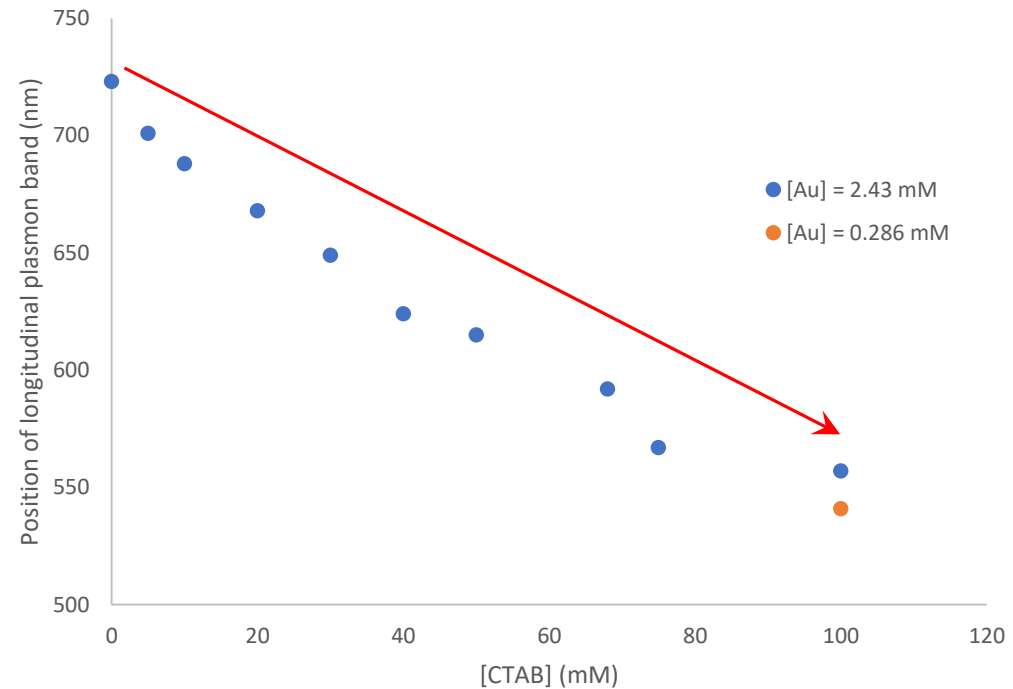
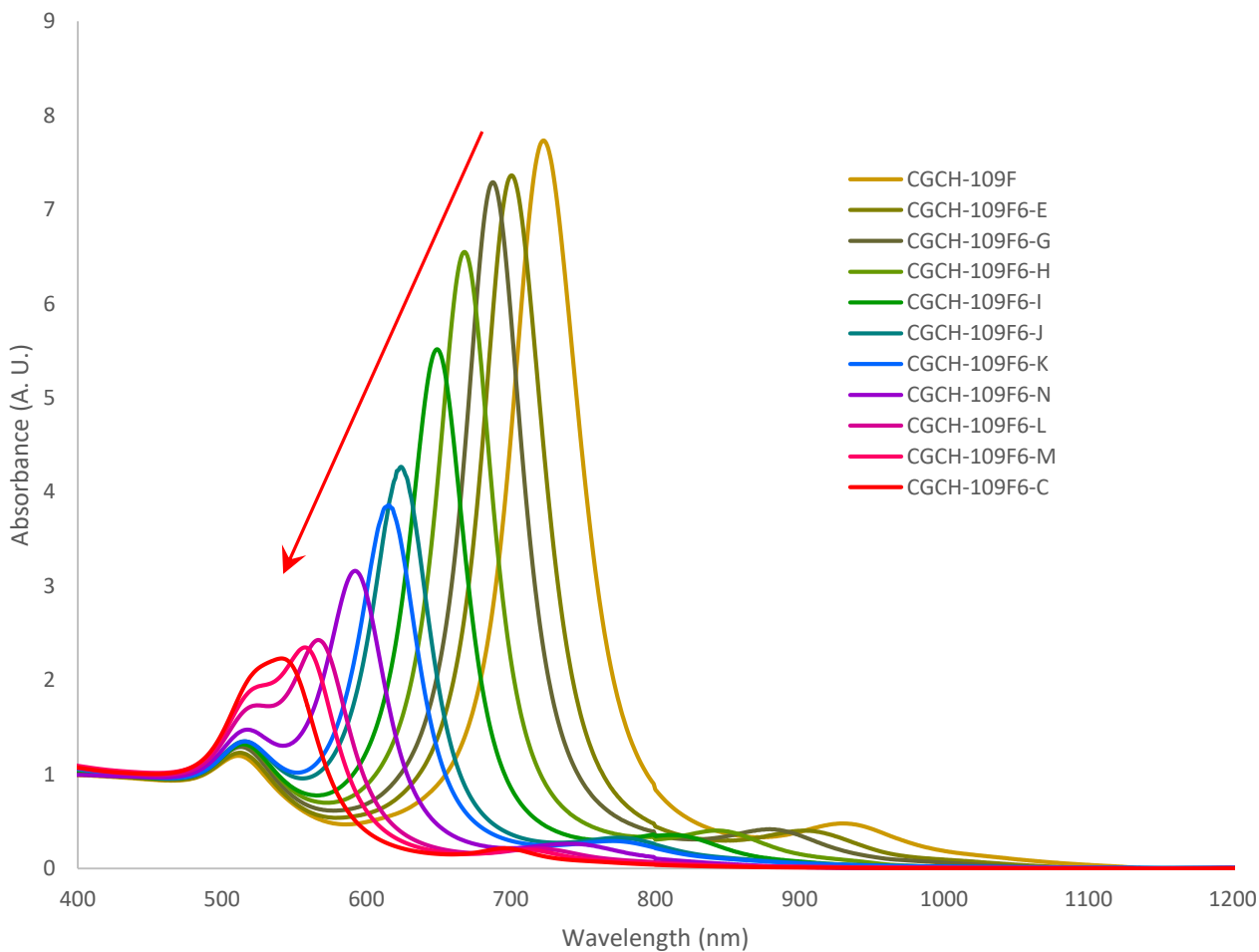


Sample	CTAC 200 mM (µL)	CTAB 200 mM (µL)	Water (µL)	V tot (µL)	[Au] (mM)	[CTAC] (mM)	[CTAB] (mM)	Time (h)
CGCH-109F							0	
CGCH-109F6-E	1000	50	950	2000	2.43	100	5	3
CGCH-109F6-G	1000	100	900	2000	2.43	100	10	7
CGCH-109F6-H	1000	200	800	2000	2.43	100	20	7
CGCH-109F6-I	1000	300	700	2000	2.43	100	30	7
CGCH-109F6-J	1000	400	600	2000	2.43	100	40	7
CGCH-109F6-K	1000	500	500	2000	2.43	100	50	7
CGCH-109F6-N	1000	680	320	2000	2.43	100	68	7
CGCH-109F6-L	1000	250	750	2000	2.43	100	75	7
CGCH-109F6-M	1000	1000	0	2000	2.43	100	100	7
CGCH-109F6-C	1000 + 2000 µL of CTAC 756 mM	7550	6450	17000	0,286	100	100	20

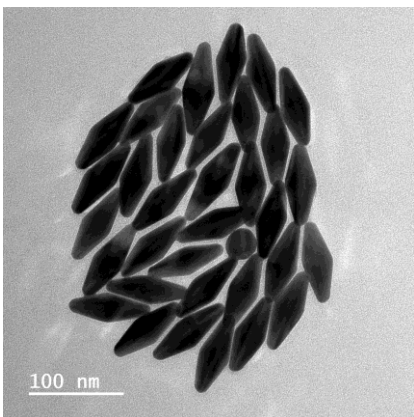
Final samples

After centrifugation, the BPs were redispersed in ≈ 10 mL of CTAC 1 mM and their spectra were recorded.
The volume of CTAC 1mM was adjusted for each sample to get a similar absorbance at 400 nm (so equal concentration of gold in each sample).

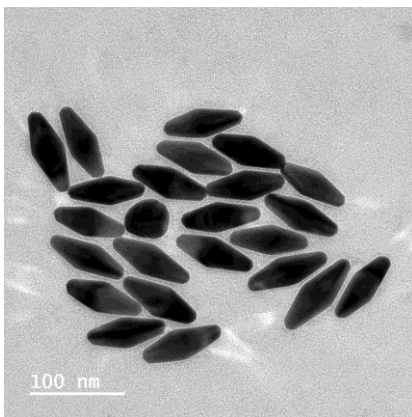
$A_{400} \approx 1$ so :
[Au] = 0.42 mM or $2.53 \cdot 10^{20}$ atoms/L
[BPs] = $2.77 \cdot 10^{-10}$ mol/L or $1.67 \cdot 10^{14}$ NPs/L



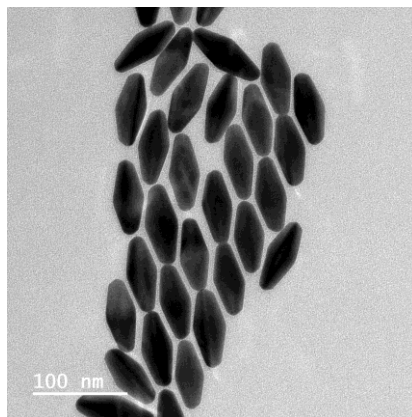
CGCH-109F



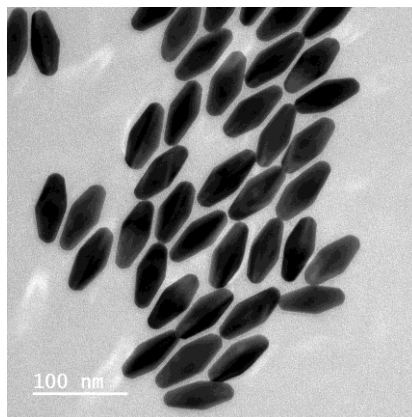
CGCH-109F6-E



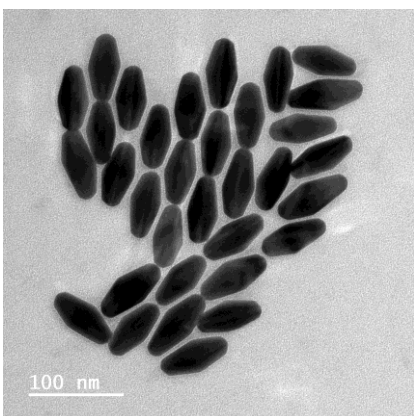
CGCH-109F6-G



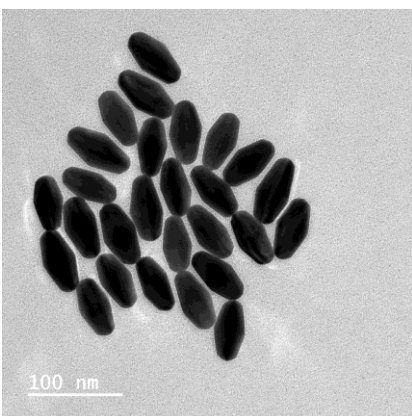
CGCH-109F6-H



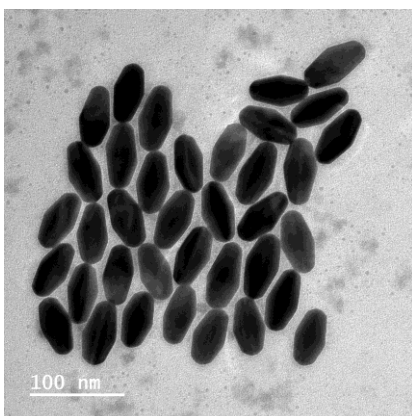
CGCH-109F6-I



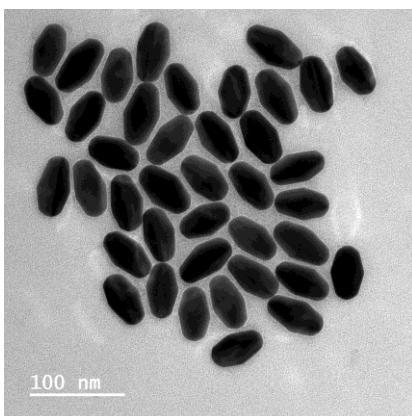
CGCH-109F6-J



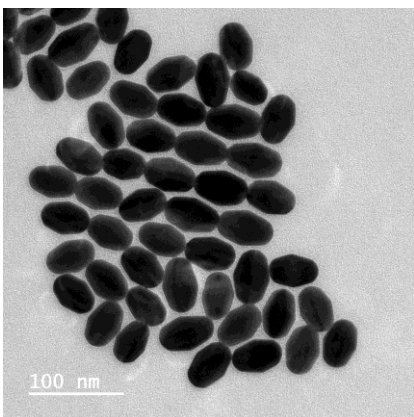
CGCH-109F6-K



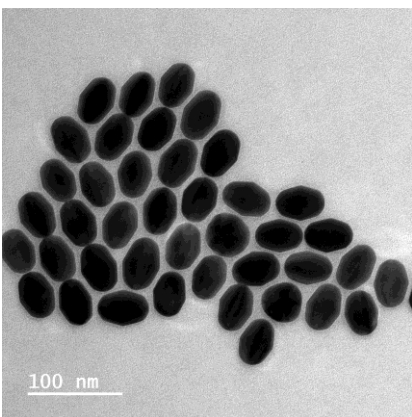
CGCH-109F6-N



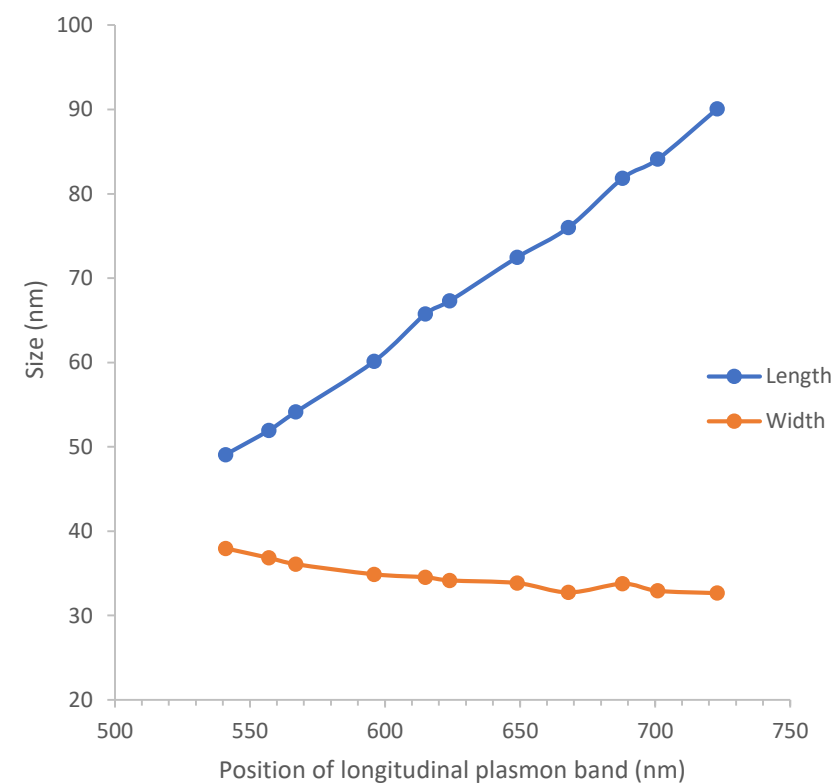
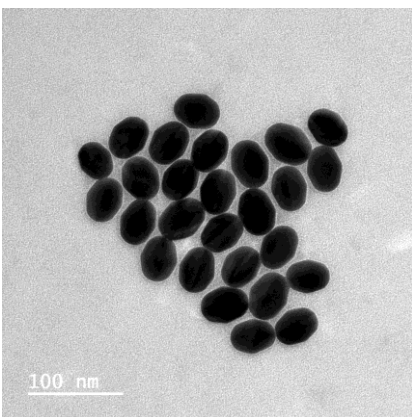
CGCH-109F6-L



CGCH-109F6-M



CGCH-109F6-C



	Experimental conditions								UV-visible spectroscopy						TEM measurements			
Sample	CTAC 200 mM (μL)	CTAB 200 mM (μL)	Water (μL)	V tot (μL)	[Au] (mM)	[CTAC] (mM)	[CTAB] (mM)	Time at 100°C (h)	λmax longitudinal (nm)	A 400nm	A 532nm	[Au] (mol/L)	[NPs] (mol/L)	E (532nm) (cm*mol*L-1)	Length (nm)	Std dev on length (nm)	Width (nm)	Std dev on width (nm)
CGCH-109F							0	0	723	0,99008931	0,93909578	4,125E-04	2,725E-10	2,902E-10	90,07	4,40	32,66	1,82
CGCH-109F6-E	1000	50	950	2000	2.43	100	5	3	701	0,99712709	0,98179431	4,155E-04	2,745E-10	2,796E-10	84,12	4,16	32,93	1,60
CGCH-109F6-G	1000	100	900	2000	2.43	100	10	7	688	1,03740004	1,04724704	4,323E-04	2,856E-10	2,727E-10	81,84	4,10	33,76	1,43
CGCH-109F6-H	1000	200	800	2000	2.43	100	20	7	668	1,03953449	1,10001808	4,331E-04	2,862E-10	2,601E-10	76,02	3,31	32,75	1,42
CGCH-109F6-I	1000	300	700	2000	2.43	100	30	7	649	1,00515297	1,11283286	4,188E-04	2,767E-10	2,486E-10	72,48	3,71	33,85	1,66
CGCH-109F6-J	1000	400	600	2000	2.43	100	40	7	624	1,07494175	1,17448413	4,479E-04	2,959E-10	2,519E-10	67,30	3,82	34,16	1,99
CGCH-109F6-K	1000	500	500	2000	2.43	100	50	7	615	1,0557487	1,18845153	4,399E-04	2,906E-10	2,445E-10	65,75	3,36	34,53	1,62
CGCH-109F6-N	1000	680	320	2000	2.43	100	68	7	596	0,99320833	1,36245462	4,138E-04	2,734E-10	2,007E-10	60,14	3,17	34,88	1,64
CGCH-109F6-L	1000	750	250	2000	2.43	100	75	7	567	1,06733573	1,72339964	4,447E-04	2,938E-10	1,705E-10	54,13	3,23	36,08	1,44
CGCH-109F6-M	1000	1000	0	2000	2.43	100	100	7	557	1,08965492	1,95712769	4,540E-04	2,999E-10	1,533E-10	51,94	2,58	36,83	1,40
CGCH-109F6-C	1000 + 2000 μL of CTAC 756 mM	7550	6450	17000	0,286	100	100	20	541	1,07027197	2,18653894	4,459E-04	2,946E-10	1,347E-10	49,08	2,34	37,95	1,66